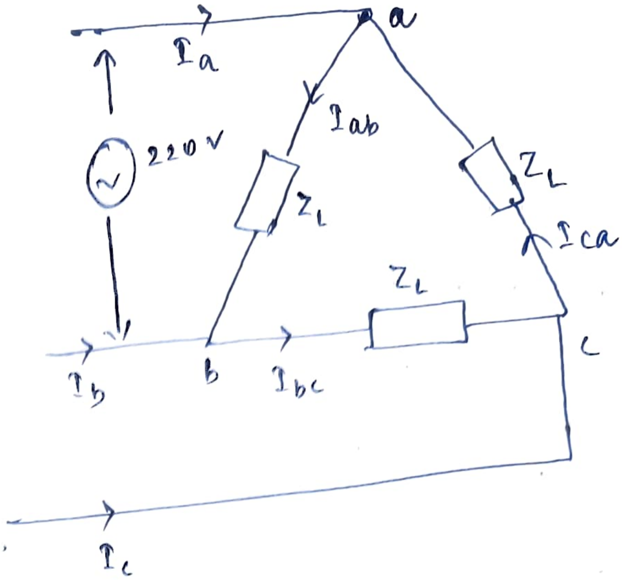
(1) Given $V_{L-L} = 220\,V$

in Δ connected load,

$\boxed{V_L = V_{ph}}$



Phase currents :-

$$I_{ab} = \frac{V_{ab}}{Z_L} = \frac{220\angle 0}{10\angle -45^\circ} = 22\angle 45^\circ\ A$$

Ily, $$I_{bc} = \frac{V_{bc}}{Z_L} = \frac{220\angle -120}{10\angle -45} = 22\angle -75^\circ\ A$$

$$I_{ca} = \frac{V_{ca}}{Z_L} = \frac{220\angle 120}{10\angle -45^\circ} = 22\angle 165^\circ\ A$$

Line current

in Δ- connection, $\boxed{I_L = \sqrt{3}\, I_{ph}\angle -30^\circ}$

so,
$$I_a = \sqrt{3}\, I_{ab}\,(\angle -30^\circ) = 38.105\angle 15^\circ\ A,$$

Ily, $$I_b = \sqrt{3}\, I_{bc}(\angle -30^\circ) = 38.105\angle -105^\circ\ A.$$

$$I_c = \sqrt{3}\, I_{ca}\,(\angle -30^\circ) = 38.105\angle 135^\circ\ A.$$

(2)

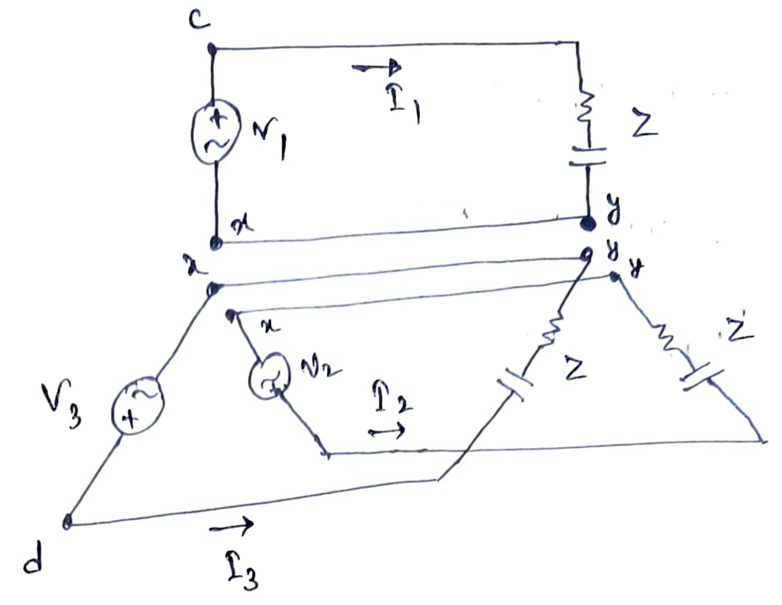
Given

$V_1 = 240\angle 90^\circ$ V

$V_2 = 240\angle -30^\circ$ V

$V_3 = 240\angle -150^\circ$ V

$Z = 40\angle -30^\circ\ \Omega$



(a) $I_1 = \dfrac{V_1}{Z} = \dfrac{240\angle 90^\circ}{40\angle -30^\circ}$ A $= 6\angle 120^\circ$ A

$I_2 = \dfrac{V_2}{Z} = \dfrac{240\angle -30^\circ}{40\angle -30^\circ} = 6$ A

$I_3 = \dfrac{V_3}{Z} = \dfrac{240\angle -150^\circ}{40\angle -30} = 6\angle -120^\circ$ A

(b) If terminals x are connected together & y are connected together

$I_{yx} = I_1 + I_2 + I_3$

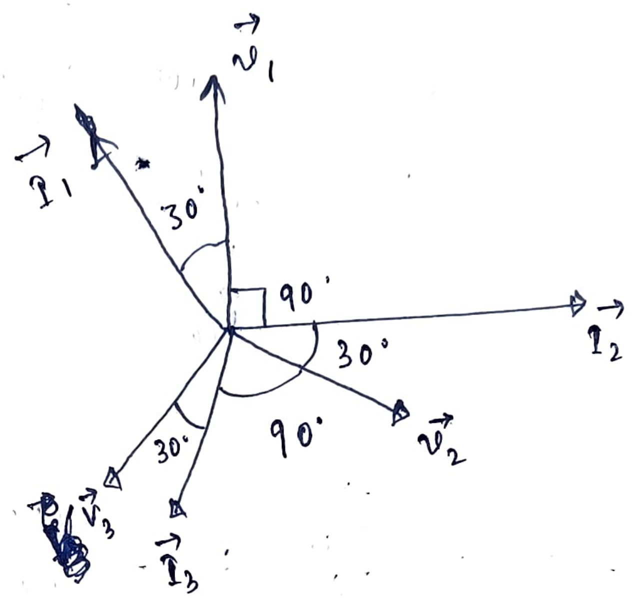
so, $I_{yx} = 6\angle 120 + 6 + 6\angle -120^\circ$

$$\boxed{I_{yx} = 0}$$

(c) $V_{cd} = V_1 - V_3 = (240\angle 90^\circ) - (240\angle -150^\circ)$

so, $\boxed{V_{cd} = 415.69\angle 60^\circ\ V}$

(d)



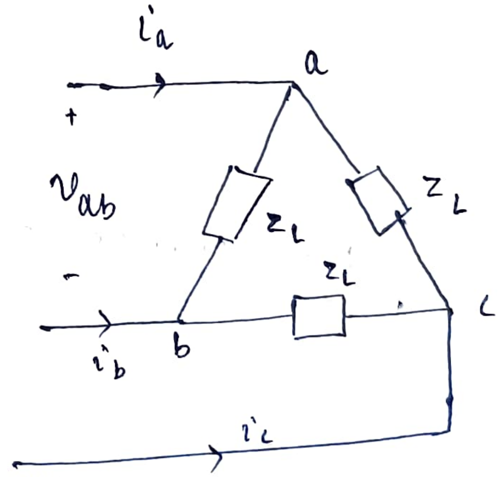
Given

(3) $V_{ab} = 240\angle 60^\circ\ V$

so, $V_{bc} = 240\angle -60^\circ\ V$

$V_{ca} = 240\angle -180^\circ\ V$

or, $240\angle 180^\circ\ V$



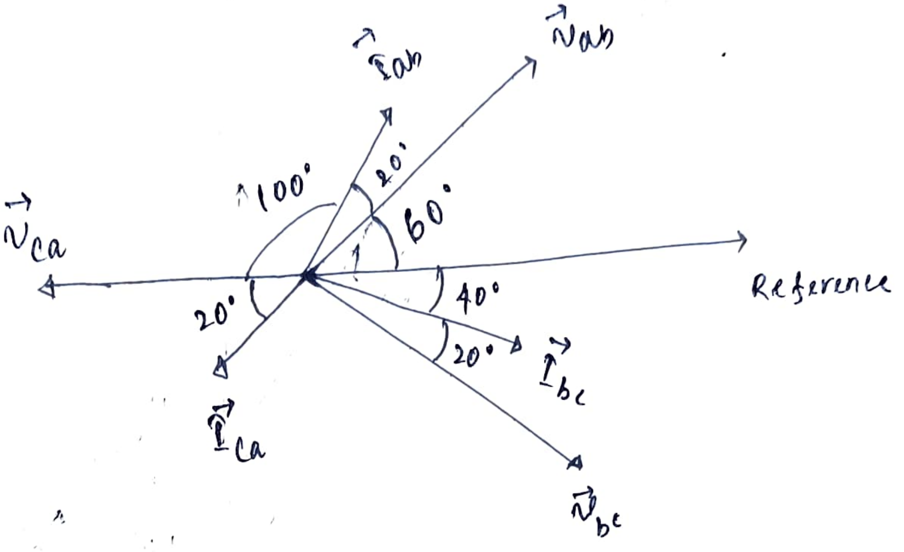
as, if we consider inductive or capacitive load, then phase angle of phase current would not exceed than ± 90° to its corresponding phase voltage

so, $I_{ca} = 6\angle 200^\circ\ A$

$I_{ab} = 6\angle 80^\circ\ A$

$I_{bc} = 6\angle -40^\circ\ A$

Phasor Diagram



(b) $Z_{ab} = Z_L = \dfrac{\vec{V}_{ab}}{\vec{I}_{ab}} = \dfrac{240 \angle 60}{6 \angle 80} = 40 \angle -20° \ \Omega$

$= (37.59 - j\,13.68)\ \Omega$

(c)

(c) $\boxed{\text{Line current} = \sqrt{3} \times \text{phase current} \times (1 \angle -30°)}$

for Δ - connection

so, $I_a = \sqrt{3} \times I_{ab} \times \angle -30° = 10.39 \angle 50°\ A$

Ily, $I_b = I_a \angle -120° = 10.39 \angle -70°\ A$

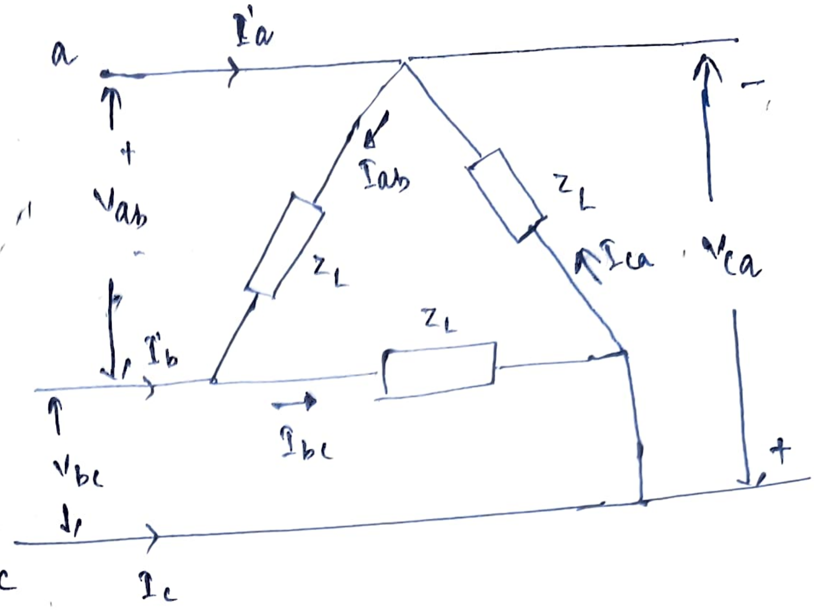
$I_c = I_b \angle -120° = 10.39 \angle -190°\ A$

(4) (a) $Z_L = 20\angle -30^\circ \ \Omega$

$V_{ab} = 240^\circ \ V$

$V_{bc} = 240\angle -120^\circ \ V$

$V_{ca} = 240\angle -240^\circ \ V$



(b) In Δ connection ; $V_{ph} = V_{line}$

<u>Phase currents</u>

$$I_{ab} = \frac{V_{ab}}{Z_L} = \frac{240}{20\angle -30^\circ} = 12\angle 30^\circ \ A$$

lly, $I_{bc} = I_{ab}\angle -120^\circ = \frac{V_{bc}}{Z_L} = 12\angle -90^\circ \ A$

$I_{ca} = I_{bc}\angle -120^\circ = I_{ab}\angle 120^\circ = \frac{V_{ca}}{Z_L} = 12\angle 150^\circ \ A$

<u>Line currents</u>

for Δ connection, $I_{Line} = \sqrt{3}\, I_{ph}(\angle -30^\circ)$

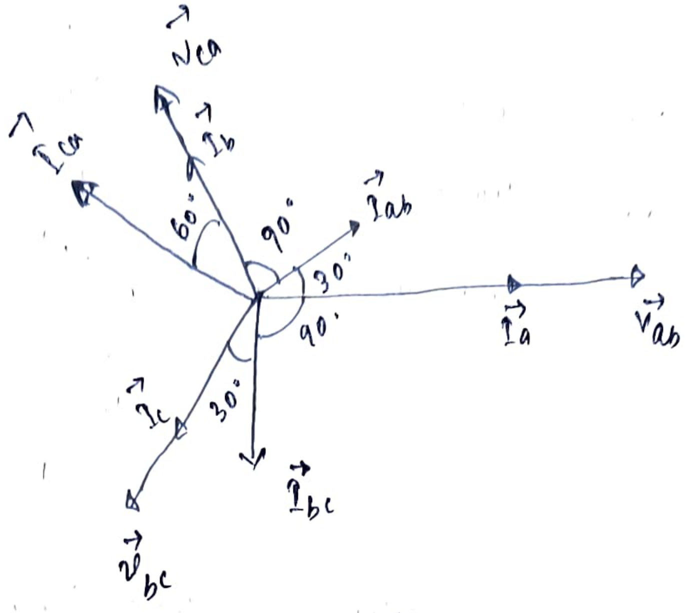
so, $I_a = \sqrt{3} \times I_{ab} \times \angle -30^\circ = 20.78 \ A$

$I_b = I_a\angle -120 = \sqrt{3} \times I_{bc} \times \angle -30^\circ = 20.78\angle -120^\circ \ A$

$I_c = I_a\angle 120 = I_b\angle -120 = \sqrt{3} \times I_{ca} \times \angle -30^\circ = 20.78\angle 120^\circ \ A$

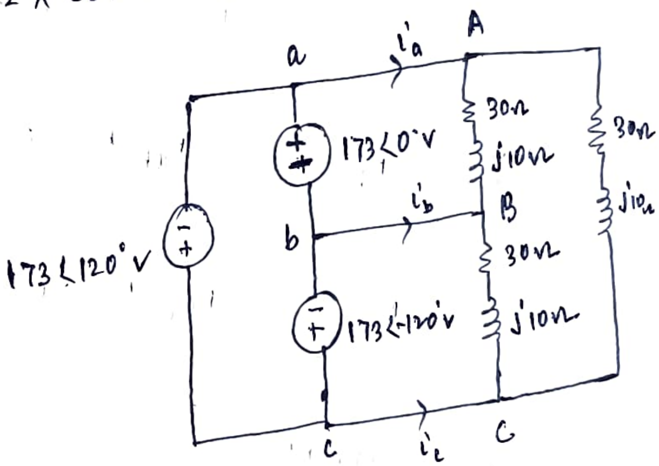
(c)

Phasor Diagram



(d) Total power $= 3 V_{ph} I_{ph} \cos\phi$

$= 3 \times 240 \times 12 \times \cos 30° = 7.48\ kW$



(5) Phase currents :-

$$I_{AB} = \frac{V_{ab}}{30+j10} = \frac{173}{30+j10}$$

$$\boxed{I_{AB} = 5.47 \angle -18.43° \ A}$$

lly, $I_{BC} = \dfrac{V_{bc}}{30+j10} = 5.47 \angle -138.43° \ A$

$I_{CA} = \dfrac{V_{ca}}{30+j10} = 5.47 \angle 101.57° \ A$

Line currents :-

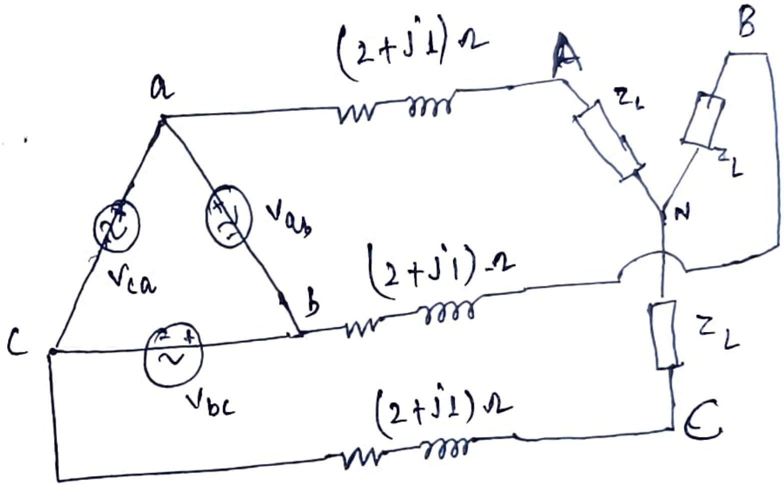
In Δ connection :- $I_L = \sqrt{3}\, I_{ph} \angle -30^\circ$

so, $I_a = 9.47 \angle -48.43^\circ$ A

$I_b = 9.47 \angle -168.43^\circ$ A

$I_c = 9.47 \angle 71.57^\circ$ A

(6) Given $Z_L = (6 + j4)\ \Omega$



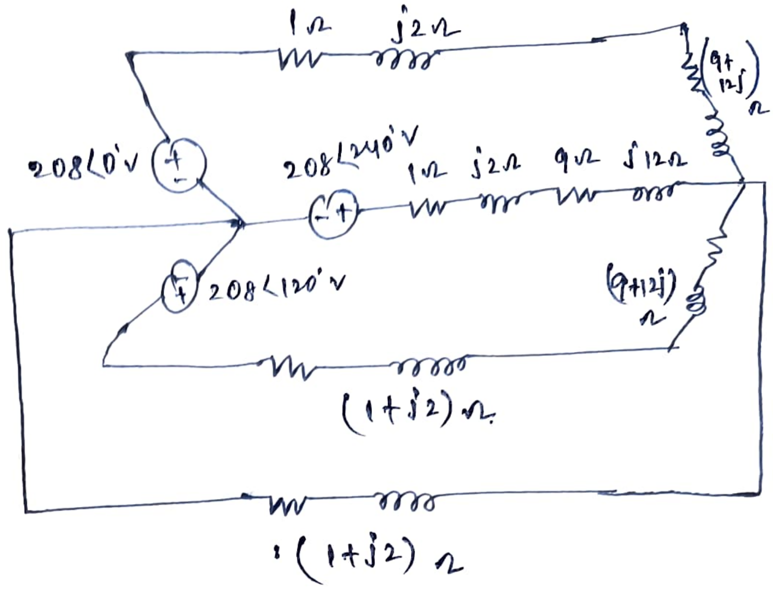
$$I_{AN} = \frac{V_{ab} \angle -30^\circ}{\sqrt{3}\,(8+j5)} = 12.73 \angle -62.005^\circ \text{ A}$$

$$V_{ph,\,load} = I_{AN} \times Z_L = 91.79 \angle -28.31^\circ \text{ V}$$

$$V_{line,\,load} = V_{ph} \times \sqrt{3} \times \angle 30^\circ$$

so, $\boxed{V_{line,\,load} = 158.99 \angle 1.68^\circ \text{ V}}$

(7)



Given configuration is star connected load with star connected source

so,

$$I_{phase} = I_{Line} = \frac{V_{Phase}}{\text{Line Impedance} + \text{Load Impedance}}$$

$$I_{ph} = \frac{208\angle 0}{10+14j} = 12.09\angle -54.46^\circ \text{ A}$$

$$V_{Load,ph} = Z_{Load} \times I_{ph} = (9+12j)(12.09\angle -54.46^\circ) \text{ V}$$

$$V_{Load,ph} = 181.35\angle -1.33 \text{ V}$$

$$\text{complex power } (S)_{Load} = 3\, V_{ph,Load} \times I_{ph}^{*}$$

$$\boxed{S_{Load} = (3946.55 + j\,5262.05) \text{ VA}}$$